國立臺南大學資訊工程學系

資料庫系統專案期末報告

Purchase-Sales-Inventory Management System of Li-Sheng-Chang Corporation

立盛昌陶瓷有限公司進銷存管理系統

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1.資訊化問題與現況

1.1 資訊化的問題與現況

目前合作的公司採用 Excel 記錄進貨及瑕疵品等相關資料，在查詢、新增、修改、刪除資料及統計商品等操作會耗費大量時間操作 Excel，加上合作公司近來年有意合作的客戶及批發品項逐年上升，因此若能將這些資訊透過更具效率的方式呈現，不但能減少操作上大量耗費時間，也能使整體上的流程更為流暢。

1.2 預期效益

為了改善及解決上述 1.1 提及問題，我們嘗試開發「進銷存管理系統」，並期望能透過此系統解決以下問題:

(A) 統計商品

(B) 減少操作上的時間成本

2.系統相關規格與說明

2.1 軟體環境

2.1.1 前端

CSS3、JavaScript、HTML5

2.1.2 後端

PHP Version 8.0.12(VSCode)、MySQL

2.2 硬體環境

Operating System: Window10 家用版(64-bit)

CPU: Intel Core i7 8750H @ 2.20GHz
RAM: 8.00GB Single-Channel Unknown @ 1330MHz(19-19-19-43)
Graphics: Intel UHD Graphics 630(ASUStek Computer Inc)
NVIDIA GeForce GTX 1050(ASUStek Computer Inc)

3.系統功能需求分析

3.1 企業規則描述

(A)可以新增商品及廠商

(B)新增進貨單時可透過下拉式選單選取商品及廠商

(C)統計每項商品剩餘數量

(D)計算每筆訂單價錢

(E)可分類職員的職權

(F)依職權限制可操作的功能

(G)記錄進貨的報銷

3.2 開發限制條件

由於合作對象與其客戶簽訂保密協定，故所有資料皆以模擬資料替代。

3.3 系統功能

(A)帳密登入登出

(B)新增廠商(廠商名稱、負責人、地址、電話)

(C)新增商品(廠商名稱、商品名稱、商品入數)

(D)顯示訂單

(E)搜尋訂單

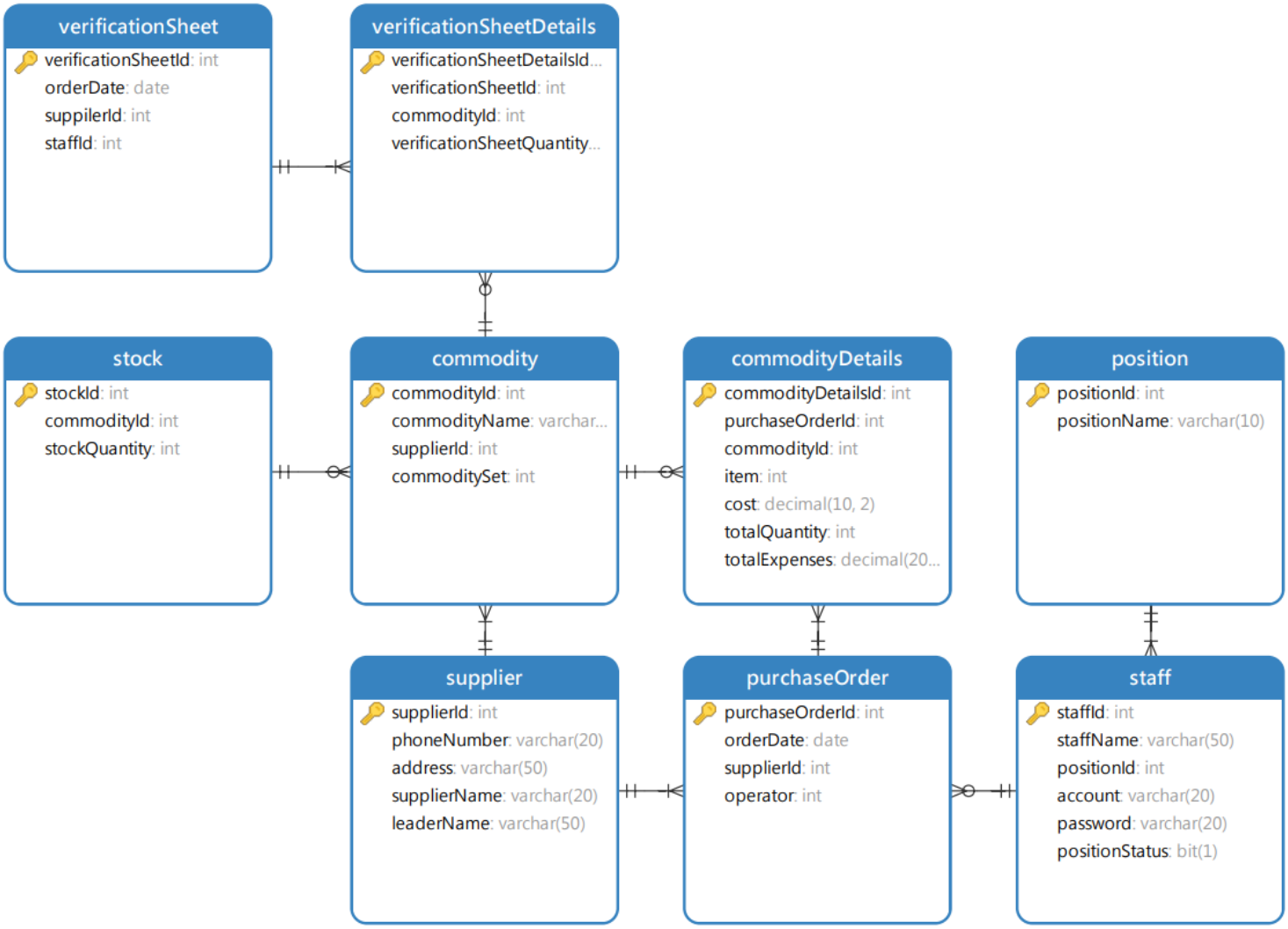
(F)新增訂單

(G)庫存資料

(H)報銷單

(I)權限管控(新增職員、編輯職員資料及狀態)

3.4 資料庫模型



4.系統介面分析

4.1 帳密登入登出

讓使用者登入使用,使用者在登入時要輸入帳號與密碼,完成之後點選登入進入系統,系統會根據資料庫的指令內容判斷該帳號與密碼是否正確並回應使用者,其對應(signIn.php)語法如下:

```
$sql = "SELECT * FROM `staff` WHERE account = '".$account."'";
```

並且可依其帳號職位別進行權限管控，對應語法如下:

```
if($data["staffId"] == 4000)
    echo "<script src='../JS/navbar.js'></script>";
else if($data["staffId"] == 4001)
    echo "<script src='../JS/navbarPersonnel.js'></script>";
else if($data["staffId"] == 4002)
    echo "<script src='../JS/navbarTreasure.js'></script>";
else if($data["staffId"] == 4003)
    echo "<script src='../JS/navbarWarehouseStaff.js'></script>";
```

4.2 新增廠商(廠商名稱、負責人、地址、電話)

讓使用者可以新增合作廠商之相關資料，對應

(addcompany.php)語法如下:

```
$sql = "INSERT INTO `supplier`(`phoneNumber`, `address`, `supplierName`, `leaderName`)
        VALUES ('$phone','$address','$companyName','$bossName')";
```

公司編號	電話號碼	地址	公司名稱	負責人名稱
7000	06-6548854	台南市中西區樹林街二段33號	惠亞瓷器股份有限公司	高文彬
7001	06-7899552	台南市東區大學路1號	現代瓷器股份有限公司	秦嘉鴻
7002	02-9985554	新北市樹林鎮東興街1號	新寶瓷器股份有限公司	林棟材
7003	02-8499256	新北市新店市寶橋路229號	原帥瓷器股份有限公司	李春淵
7012	0355555555	asddd	台南大學	aaa

1

4.3 新增商品(廠商名稱、商品名稱、商品入數)

讓使用者可以新增合作商品之相關資料，對應 (addproduct.php)

語法如下:

```
$sql = "INSERT INTO `commodity`(`commodityName`, `supplierId`, `commoditySet`)
        VALUES ('$productName','$ID','$prouductSet')";
```

透過下拉式選單進行對廠商名稱進行新增，其下拉式選單語法如下:

```
while($row = mysqli_fetch_assoc($result)){
    $i++;
    $name = $row['supplierName'];
    echo "<option value=$name>";
    echo $name;
    echo "</option>";
    }
```

4.4 顯示訂單

影片說明:

https://drive.google.com/file/d/1ujqXNvPAyyzYZwlCG1z-BOTNVtz9yRBJ/view?usp=sharing

(A) purchase.php

本頁面只要是查看進貨單的頁面，可以對整個單據選擇刪除或進行查詢，其對應查詢語法如下:

另外刪除是透過傳值至 URL 再由另一個對應之頁面取值來進行刪除動作，以下為傳值至 URL 語法:

```
$sql = "SELECT  `orderDate`,purchaseorder.`purchaseOrderId`,`supplierName`,`staffName`
        FROM `commodity`
        INNER JOIN `commoditydetails`ON `commodity`.commodityId = `commoditydetails`.commodityId
        INNER JOIN `purchaseorder` ON `purchaseorder`.purchaseOrderId = `commoditydetails`.purchaseOrderId
        INNER JOIN `supplier` ON `purchaseorder`.supplierId = supplier.supplierId
        INNER JOIN `staff` ON `purchaseorder`.operator = `staff`.staffId
        GROUP BY purchaseorder.`purchaseOrderId`
        ORDER BY purchaseorder.`purchaseOrderId`";
```

```
while ($rows = mysqli_fetch_array($result, MYSQLI_BOTH) and $j <= $records_per_page) {

    echo "<tr>";
    for ( $i = 0; $i<= $total_fields-1; $i++ ){
        echo "<td align='center' valign='middle'>".$rows[$i]."</td>";

    }
    echo "<td style='width:005px' ><a href='modifyOrder.php?action=Del&id=";
        echo $rows[1]."'><b align='center'>刪除單據</b></td>";

    echo "<td style='width:005px' ><a href='commodityDetails.php?action=edit&id=";
        echo $rows[1]."'><b align='center'>查詢明細</b></td>";
        echo "</tr>";
    echo "</tr>";
    $j++;

}
```

(B) commodityDetails.php

本頁面主要可以對每項單據進行編輯&查詢，而編輯及查詢等功能語法也是透過傳特定值至 URL 再進行刪除等查詢等，本頁面主要語法為顯示查詢的結果，其對應語法如下:

```
$sql = "SELECT purchaseorder.`purchaseorderId`, commodity.`commodityName`,commodity.`commoditySet`
            , commodityDetails.`item`,commodityDetails.`cost`
            ,(commodity.`commoditySet` * commodityDetails.`item` * commodityDetails.`cost`)
            ,commoditydetails.`commodityId`
        FROM purchaseorder
        JOIN commoditydetails  ON  purchaseorder.purchaseOrderId = commoditydetails.purchaseOrderId
        JOIN supplier ON purchaseOrder.supplierId = supplier.supplierId
        JOIN staff ON purchaseOrder.operator = staff.staffId
        JOIN commodity ON commodityDetails.commodityId = commodity.commodityId
        WHERE purchaseOrder.purchaseOrderId = $id";
```

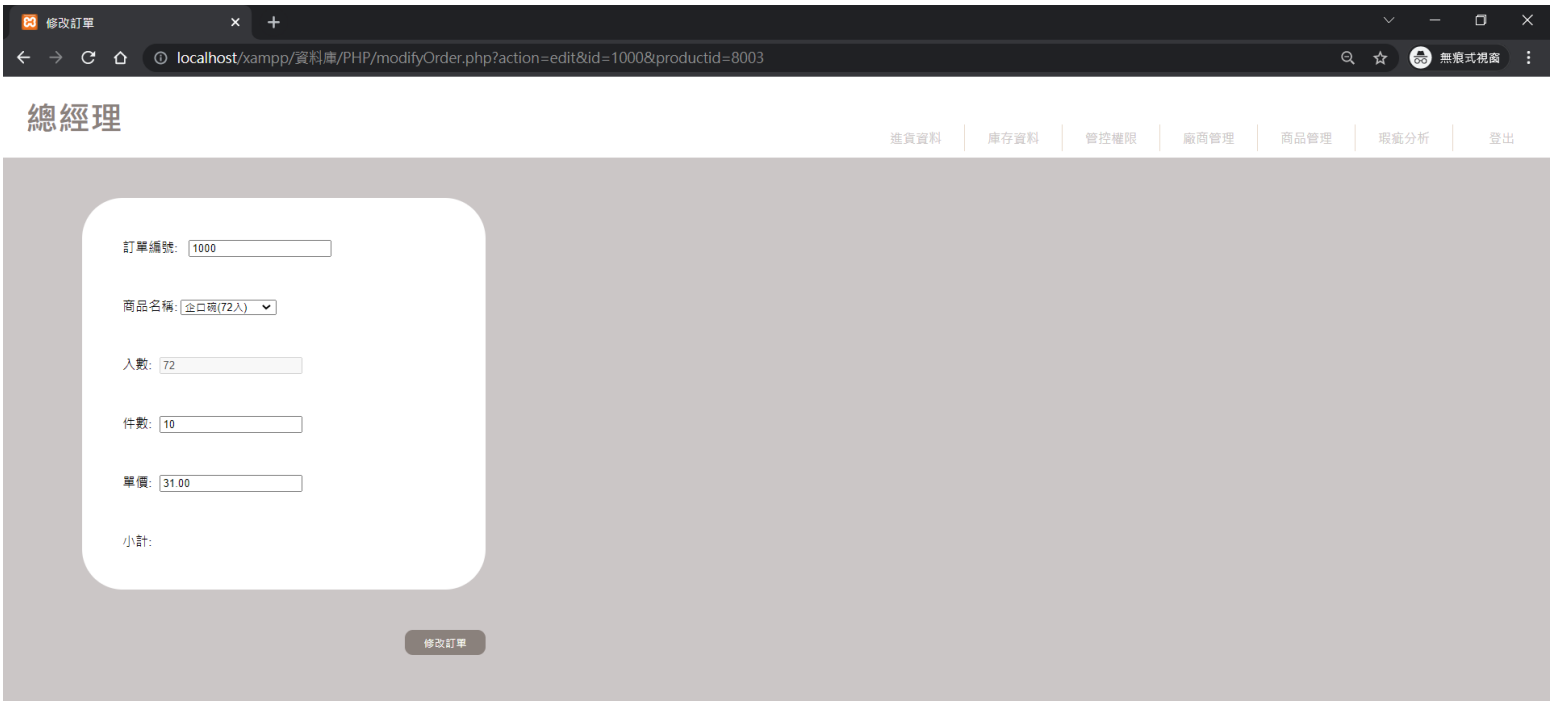
(C) modifyOrder.php

本頁面主要負責前面(A)(B)兩頁面所點選特定功能撰寫對應

功能之語法，刪除、編輯、更新，語法如下:

```php
$id = $_GET["id"];  // 取得URL參數的編號
$productid = $_GET["productid"];
$action = $_GET["action"];  // 取得操作種類

switch ($action) {
    case "update": // 更新操作
        $commodityName = $_POST["commodityName"];
        $commoditySet = $_POST["commoditySet"];
        $item = $_POST["item"];
        $cost = $_POST["cost"];
        $totalQuantity = $commoditySet * $item;
        $totalExpenses = $commoditySet * $item * $cost;

        $sql3 = "UPDATE commoditydetails
                    LEFT JOIN commodity ON commoditydetails.commodityId = commodity.commodityId
                    SET commoditydetails.commodityId = '$commodityName',
                    item = '$item',
                    cost = '$cost',
                    totalQuantity = '$totalQuantity',
                    totalExpenses = '$totalExpenses'
                    WHERE commoditydetails.purchaseOrderId = '$id'";

        mysqli_query($link, $sql3);  // 執行SQL指令
        header("Location: commodityDetails.php?id=$id"); // 轉址
    break;

    case "del":
        $sql = "DELETE FROM commoditydetails  WHERE commoditydetails.purchaseOrderId = $id AND commoditydetails.commodityId = $productid";
        mysqli_query($link, $sql);
        header("Location: purchase.php"); // 轉址
    break;
```

```php
case "del":
    $sql = "DELETE FROM commoditydetails  WHERE commoditydetails.purchaseOrderId = $id AND commoditydetails.commodityId = $productid";
    mysqli_query($link, $sql);
    header("Location: purchase.php"); // 轉址
break;

case "Del":  //用於刪除可能產生外部鍵問題的頁面
    $sql1 = "DELETE FROM commoditydetails  WHERE commoditydetails.purchaseOrderId = $id";
    mysqli_query($link, $sql1);
    $sql2 = "DELETE FROM purchaseorder WHERE purchaseorder.purchaseOrderId = $id";
    mysqli_query($link, $sql2);

    header("Location: purchase.php"); // 轉址
break;

case "edit":   // 編輯操作
    $sql = "SELECT * FROM purchaseorder JOIN commoditydetails  ON  purchaseorder.purchaseOrderId = commoditydetails.purchaseOrderId
                JOIN supplier ON purchaseOrder.supplierId = supplier.supplierId
                JOIN staff ON purchaseOrder.operator = staff.staffId
                JOIN commodity ON commodityDetails.commodityId = commodity.commodityId
                WHERE purchaseOrder.purchaseOrderId = $id AND commoditydetails.commodityId = $productid";

    $result = mysqli_query($link, $sql); // 執行SQL指令
    $row = mysqli_fetch_assoc($result); // 取回記錄
    $orderDate = $row["orderDate"];
    $purchaseOrderCode = $row["purchaseOrderId"];
    $supplierName = $row["supplierName"];
    $staffName = $row["staffName"];
    $commodityName = $row["commodityName"];
    $commoditySet = $row["commoditySet"];
    $item = $row["item"];
    $cost = $row["cost"];
    $totalExpenses = $row["totalExpenses"];
```


修改訂單
localhost/xampp/資料庫/PHP/modifyOrder.php?action=edit&id=1000&productid=8003
無痕式視窗
總經理
進貨資料
庫存資料
管控權限
廠商管理
商品管理
瑕疵分析
登出
訂單編號: 1000
商品名稱: 企口碗(72入)
入數: 72
件數: 10
單價: 31.00
小計:
修改訂單

4.5 搜尋訂單

影片說明:

https://drive.google.com/file/d/1h7gl2DNQa-2O3f_mHet2RmwVOL4ZuKtf/view?usp=sharing

本頁面(search.php)是用來進行對資料庫的搜尋，採用 OR 邏輯的方式進行搜尋，其對應語法如下:

```
$sql1 = "SELECT DISTINCT * FROM `purchaseorder`
            inner join  `commoditydetails` ON purchaseorder.purchaseOrderId = commoditydetails.purchaseOrderId
            inner join  `commodity` ON commoditydetails.commodityId=commodity.commodityId
            inner join  `supplier` ON purchaseorder.supplierId = supplier.supplierId
            inner join  `staff` ON purchaseorder.operator = staff.staffId
            WHERE orderDate = '".$_date."' OR commodityName ='".$_productName."' OR supplierName = '".$_companyName."' OR staffName ='".$_staffName."'" ;
```

4.6 新增訂單

本頁面(addOrder.php)可使使用者自行新增進貨單的資料，其對應的語法如下:

```
$sql = "INSERT INTO `purchaseorder`(`orderDate`,`purchaseOrderId`,`supplierId`,`operator`)
VALUES('$orderDate','$purchaseOrderCode','$supplierName','$staffName')";
```

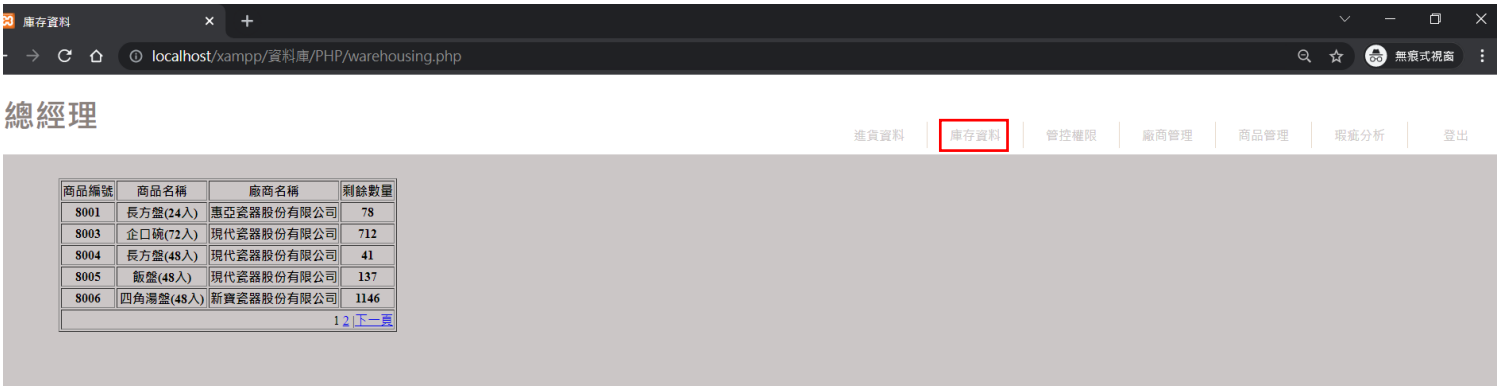
根據下拉式選單所選擇的項目進行新增資料，下拉式選單對應語法如下:

```
<?php
echo "<option value>請選擇</option>";
$sql = "SELECT * FROM commodity WHERE supplierId = $c"; //從company某個table選擇所有資料
$result = mysqli_query($link,$sql);
$i=0;
while($row = mysqli_fetch_assoc($result)){
    $i++;
    $commodityId=$row['commodityId'];
    $commoditySet=$row['commoditySet'];
    echo "<option value=$commodityId>";
    echo $row['commodityName']; //商品名稱
    echo "</option>";
}
?>
```

4.7 庫存資料

本頁面(warehousing.php)是透過查詢語法對資料庫進行查詢商品相關明細數量進行運算並顯示對應的商品，其語法如下圖:

```
$sql = "SELECT stock.`commodityId`,commodity.`commodityName`,supplier.`supplierName`
        , SUM(commoditydetails.`totalQuantity`) - SUM(verificationsheetdetails.`verificationSheetQuantity`) AS stockQuantity
        FROM `stock`
        LEFT JOIN `commodity` ON stock.commodityId = commodity.commodityId
        LEFT JOIN `supplier` ON commodity.supplierId = supplier.supplierId
        LEFT JOIN `commoditydetails` ON stock.`commodityId` = commoditydetails.`commodityId`
        LEFT JOIN `verificationsheetdetails` ON `stock`.`commodityId` = verificationsheetdetails.`commodityId`
        GROUP BY stock.`commodityId`,commodity.`commodityName`
        HAVING (stockQuantity) >0
        ORDER BY stock.`commodityId` ASC";
```



4.8 報銷單

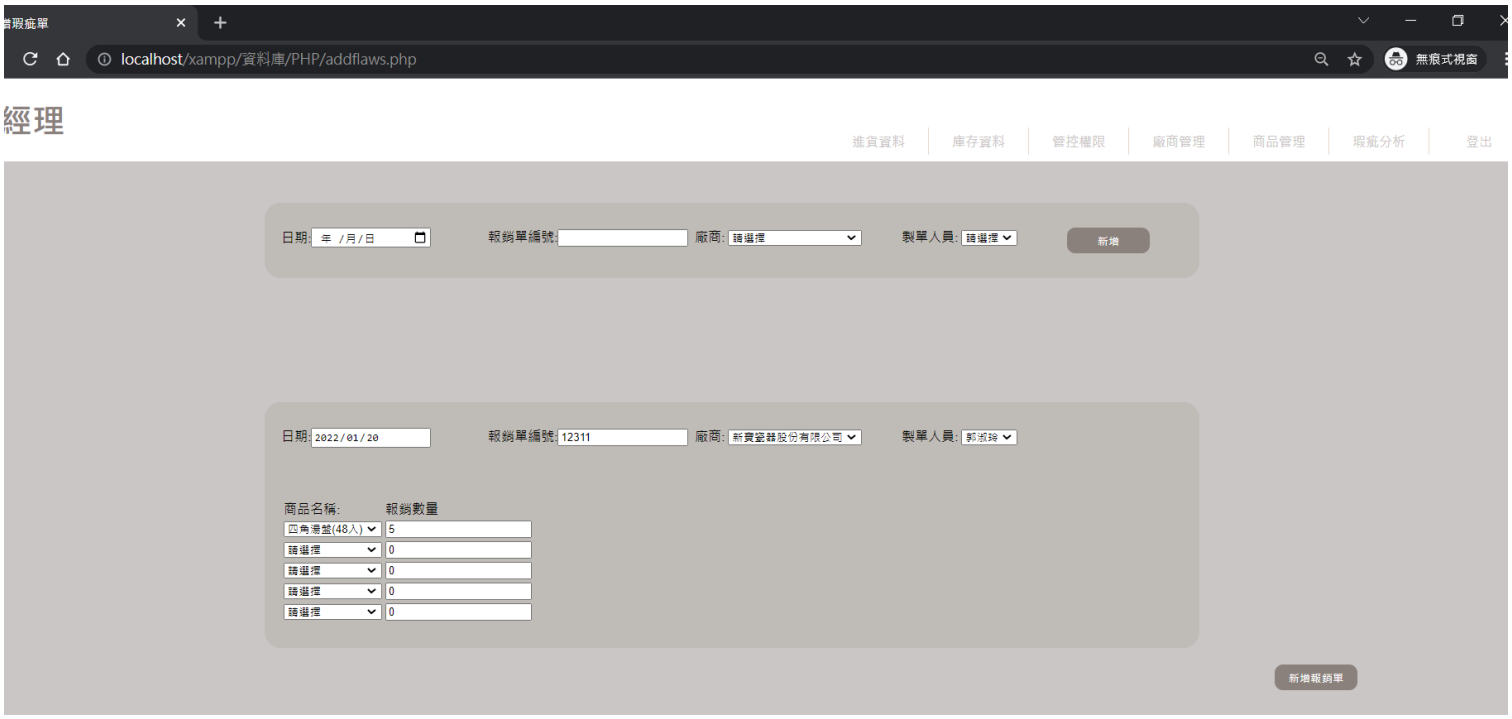
影片說明:

https://drive.google.com/file/d/1qU-vmm1PXzq1SYyPETtuE2PGI8siBgjU/view?usp=sharing

(A) addflaws.php

新增報銷之相關資料，其對應語法如下:

```
$sql = "INSERT INTO `verificationsheet`(`orderDate`,`verificationSheetId`,`suppilerId`,`staffId`)
        VALUES('$orderDate','$verificationSheetId','$supplierName','$staffName')";
```



(B) flaws.php

顯示報銷的單據，可以刪除整筆報銷單單據或查詢，其新增後所查詢到的報銷單資料，其語法如下:

```
$sql = "SELECT  `orderDate`,`verificationSheetId`,`supplierName`,`staffName`
        FROM `verificationSheet`
        INNER JOIN `supplier` ON `verificationSheet`.suppilerId = supplier.supplierId
        INNER JOIN `staff` ON `verificationSheet`.staffId = `staff`.staffId
        GROUP BY verificationSheet.`verificationSheetId`
        ORDER BY verificationSheet.`verificationSheetId`";
```

(C) flawsDetails.php

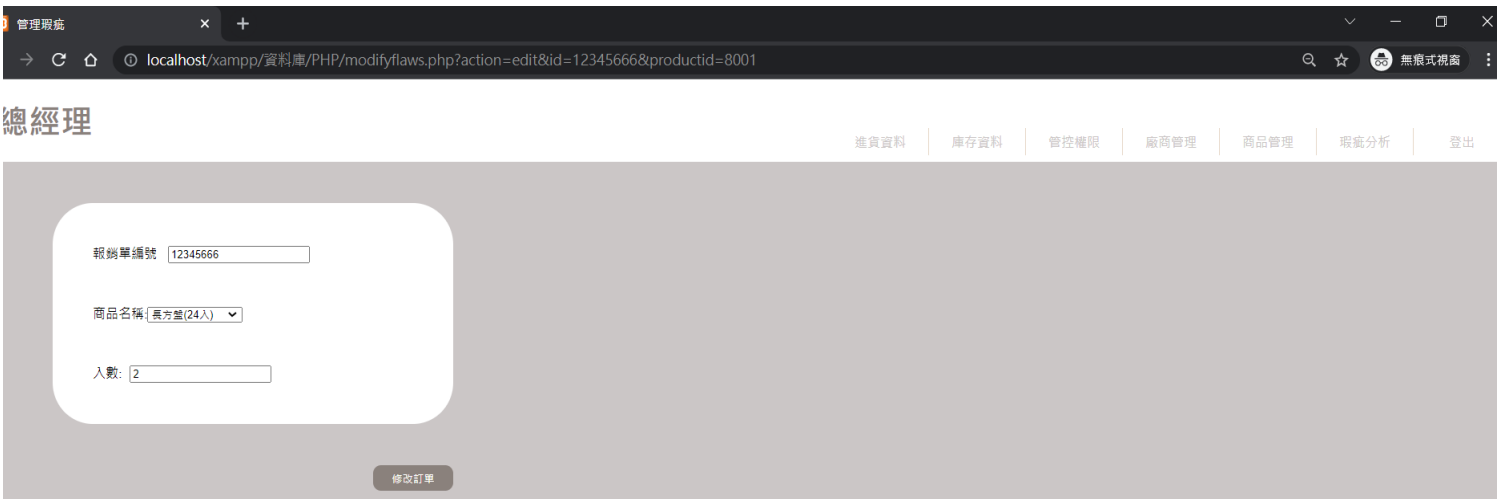
依報銷單據號碼查詢明細裡是否存在多筆記錄在同一張單號，其對應語法如下:

```
$sql = "SELECT verificationsheet.`verificationSheetId`, commodity.`commodityName`
            , verificationSheetdetails.`verificationSheetQuantity`, commodity.`commodityId`
        FROM verificationsheet
        JOIN verificationSheetdetails
            ON  verificationsheet.verificationSheetId = verificationSheetdetails.verificationSheetId
        JOIN supplier
            ON verificationsheet.suppilerId = supplier.supplierId
        JOIN staff
            ON verificationsheet.staffId = staff.staffId
        JOIN commodity
            ON verificationSheetdetails.commodityId = commodity.commodityId
        WHERE verificationsheet.verificationSheetId = $id";
```

(D) modifyflaws.php

針對某一筆報銷單進行整體單據刪除或對某一筆報銷單據之報銷明細中的某一筆特定資料進行編輯及刪除，其對應語法如下：

```php
switch ($action) {
    case "update": // 更新操作
        $commodityName = $_POST["commodityName"];
        $verificationSheetQuantity = $_POST["verificationSheetQuantity"];

        $sql3 = "UPDATE verificationSheetdetails
                SET verificationsheetdetails.commodityId = '$commodityName', verificationSheetdetails.verificationSheetQuantity = '$verificationSheetQuantity'
                WHERE verificationSheetdetails.verificationSheetId = $id AND verificationSheetdetails.commodityId = $productid";
        mysqli_query($link, $sql3);  // 執行SQL指令
        header("Location: flawsDetails.php?id=$id"); // 轉址
    break;

    case "del":
        $sql = "DELETE FROM verificationSheetdetails
                WHERE verificationSheetdetails.verificationSheetId = $id AND verificationSheetdetails.commodityId = $productid";
        mysqli_query($link, $sql);
        header("Location: flaws.php"); // 轉址
    break;

    case "Del":
        $sql1 = "DELETE FROM verificationsheetdetails  WHERE verificationsheetdetails.verificationSheetId = $id";
        mysqli_query($link, $sql1);
        $sql2 = "DELETE FROM verificationsheet WHERE verificationsheet.verificationSheetId = $id";
        mysqli_query($link, $sql2);

        header("Location: flaws.php"); // 轉址
    break;

    case "edit":   // 編輯操作
        $sql = "SELECT * FROM verificationsheet
                JOIN verificationSheetdetails  ON  verificationsheet.verificationSheetId = verificationSheetdetails.verificationSheetId
                JOIN supplier ON verificationsheet.suppilerId = supplier.supplierId
                JOIN staff ON verificationsheet.staffId = staff.staffId
                JOIN commodity ON verificationSheetdetails.commodityId = commodity.commodityId
                WHERE verificationsheet.verificationSheetId = $id AND verificationSheetdetails.commodityId = $productid";
```



4.9 權限管控(新增職員、編輯職員資料及狀態)

影片說明:

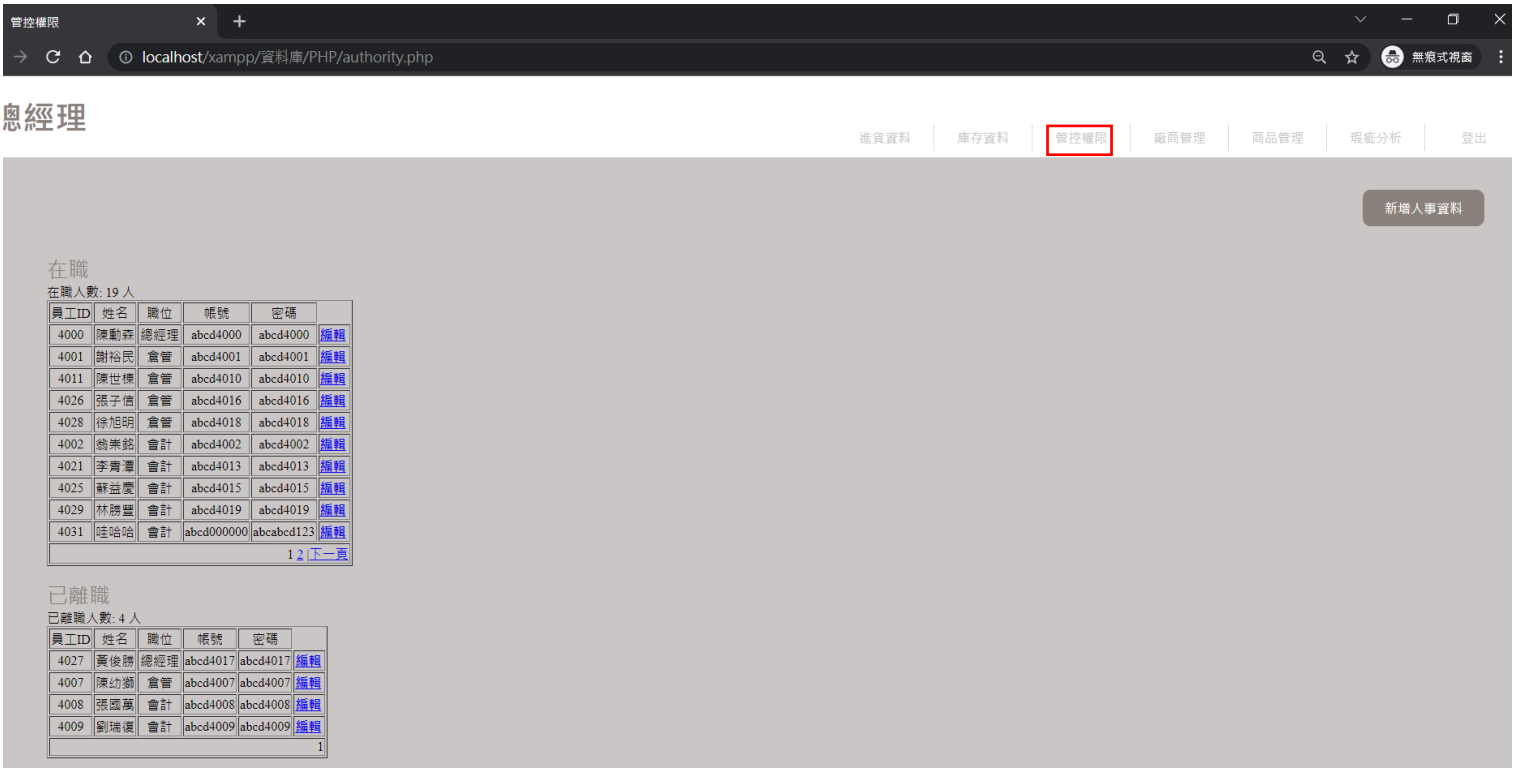
https://drive.google.com/file/d/1zk_Wv16XQh0YaO5QF_Z312TXm81H5cjp/view?usp=sharing

(A) authority.php

本頁面分為在職跟離職，透過抓取狀態值來顯示該筆資料所在的狀態欄位，其對應的語法共兩個(1:在職 0:離職):

```
$sql = "SELECT staffId,staffName,positionName,account,password
        FROM staff INNER JOIN position ON staff.positionId = position.positionId
        Where positionStatus= 1";
```

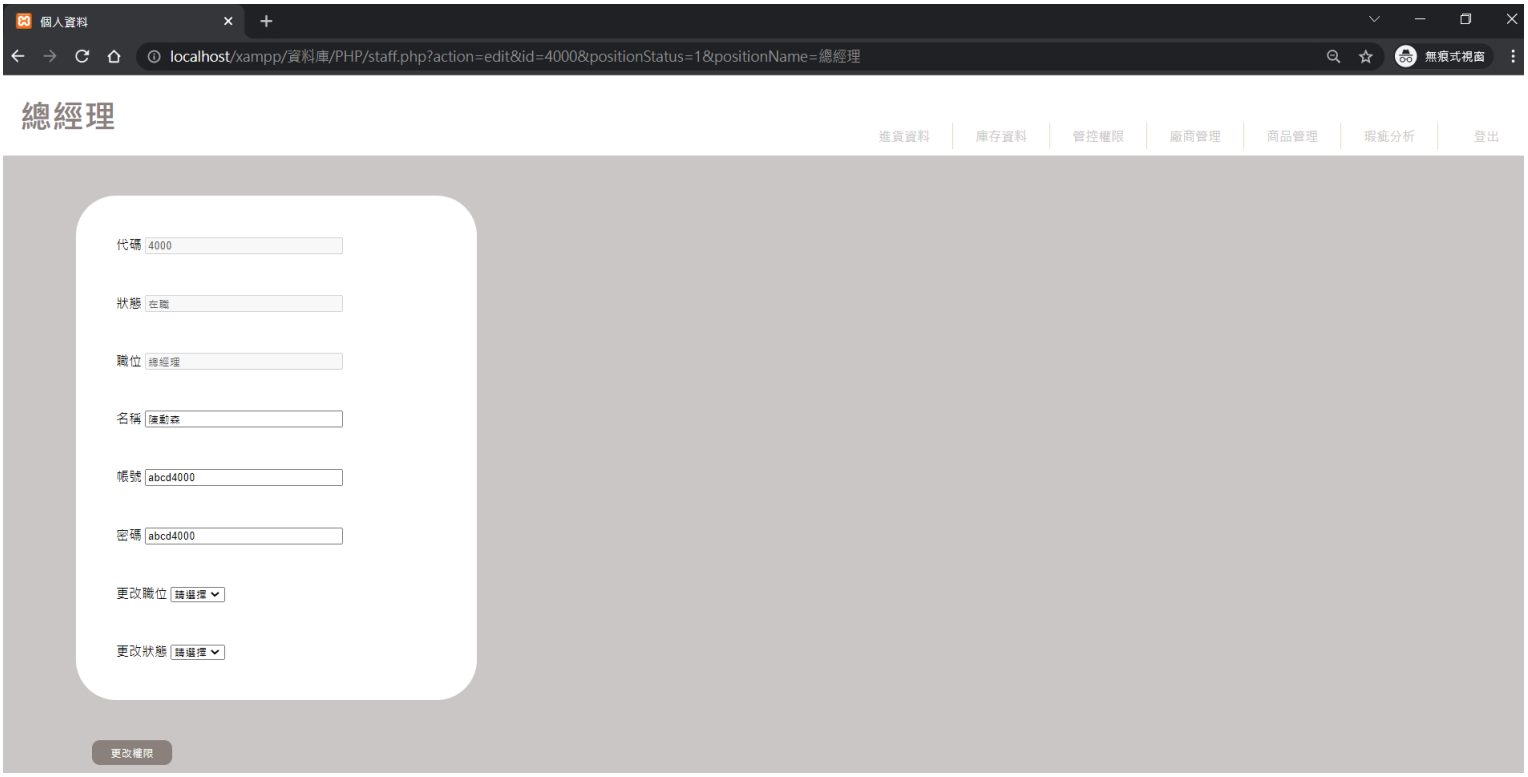
```
$sql = "SELECT staffId,staffName,positionName,account,password
        FROM staff INNER JOIN position ON staff.positionId = position.positionId
        Where positionStatus= 0";
```



(B) staff.php

本頁面為修改員工資訊的頁面，透過 authority.php 將特定值傳入至 URL 再將取值後去更改對應的員工資訊、狀態、帳號及密碼...等，其對應的語法如下:

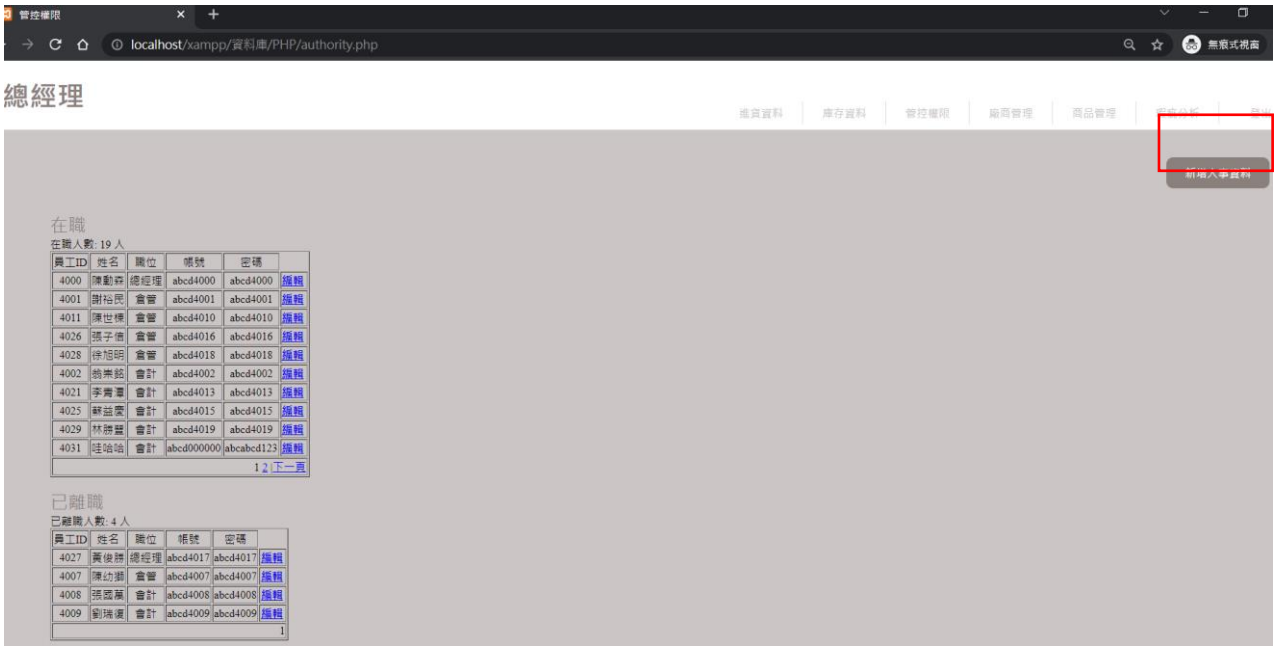
```php
switch ($action) {
  case "update": // 更新操作
    $staffID = $_POST["staffID"];
    $staffName = $_POST["staffName"];
    $staffPosition = $_POST["staffPosition"];
    $staffAccount = $_POST["staffAccount"];
    $staffPassword = $_POST["staffPassword"];
    $positionStatus = $_POST["positionStatus"];

    $sql = "UPDATE staff JOIN position ON staff.positionId = position.positionId
            SET  staffName = '$staffName' ,positionName = '$staffPosition', positionStatus='$positionStatus'
            ,account = '$staffAccount' , password = '$staffPassword'
            WHERE staffId = $id";
    mysqli_query($link, $sql);  // 執行SQL指令

    header("Location: authority.php"); // 轉址
    break;

  case "edit":   // 編輯操作
    $sql = "SELECT * FROM staff JOIN position  ON  staff.positionId = position.positionId WHERE staffId = $id";
    $result = mysqli_query($link, $sql); // 執行SQL指令
    $row = mysqli_fetch_assoc($result); // 取回記錄
    $staffID = $row["staffId"];
    $staffName = $row["staffName"];
    $staffAccount = $row["account"];
    $staffPassword = $row["password"];
// 顯示編輯表單
```

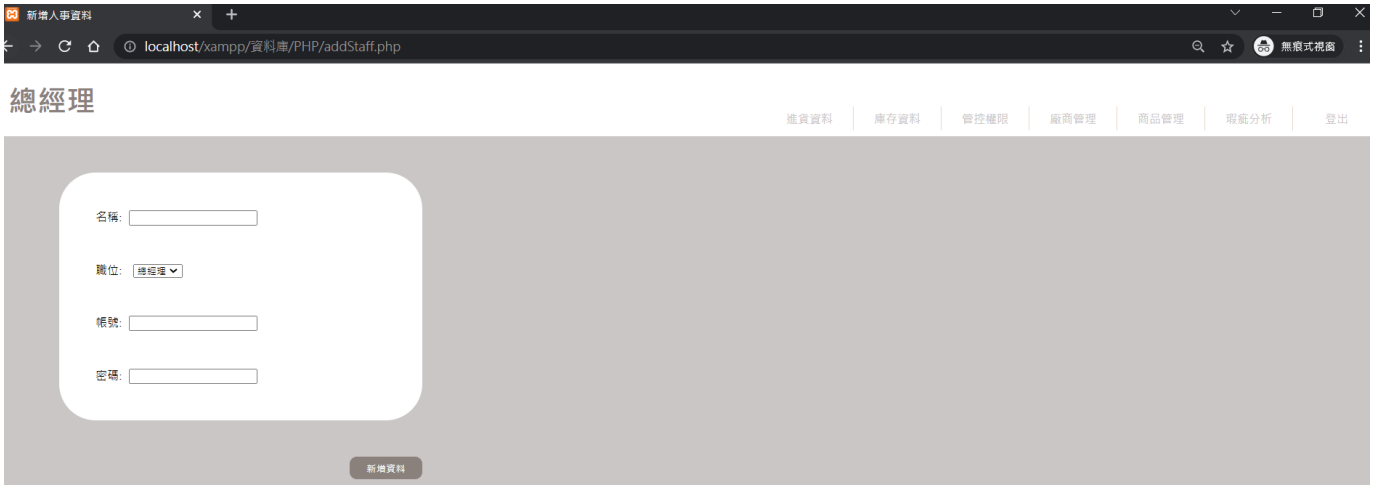


(C) addStaff.php

本頁面為新增員工資料，其對應新增語法如下圖:

```php
if ($staffName != "" && $positionName != "" && $account != "" && $password != "") {
  $sql = "INSERT INTO `staff` (`staffName`,`positionId`,`account`,`password`,`positionStatus`)
          VALUES('$staffName','$positionName','$account','$password','$positionStatus')";

  if ( mysqli_query($link, $sql)) { // 執行SQL指令
    echo "<font color=red>新增聯絡資料成功!";
    echo "</font><br/>";
  }
  else{
    echo "<font color=red>新增聯絡資料失敗!";
    echo "</font><br/>";
  }
}
```





5.系統非功能性需求

5.1 權限管控

影片：

https://drive.google.com/file/d/15gsBbE64wppns9V7PUn554fKniOmcXx7/view?usp=sharing

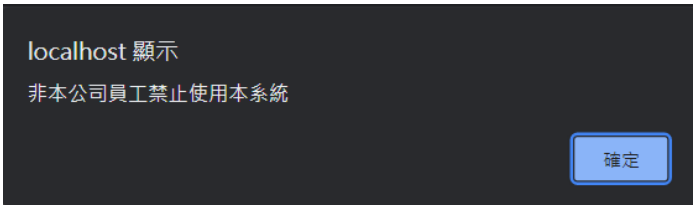
不同的職位可操作的功能不同，功能圖如下：

可對職員可以進行停權動作：

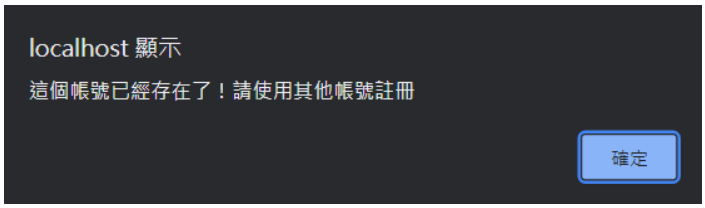
5.2 帳號登入判斷&安全性需求

若本公司員工登入的帳號密碼錯誤:

若為非本公司員工登入:



帳號若重複註冊:



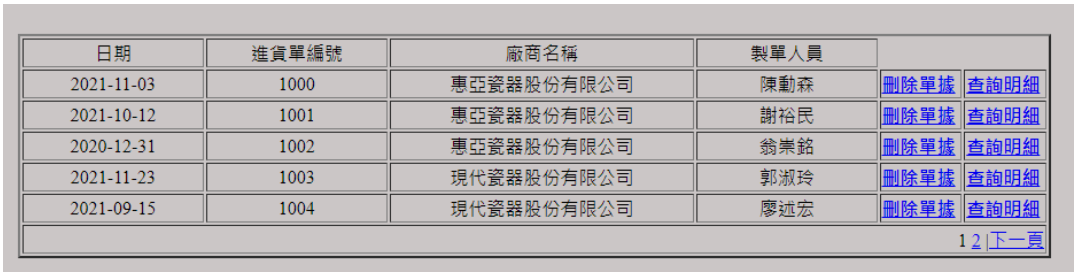
總經理

5.3 資源分配

影片：

https://drive.google.com/file/d/1SAhPyWi6zwIkn8ce59AFc2oavMqIHfU0/view?usp=sharing

每一筆資料在資料庫中都有限定資料的大小，避免分配過多空間，造成記憶體的浪費。

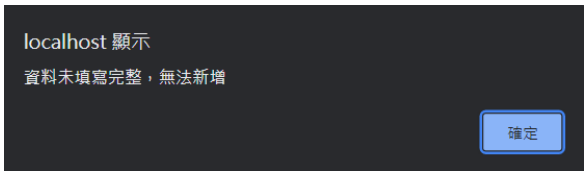


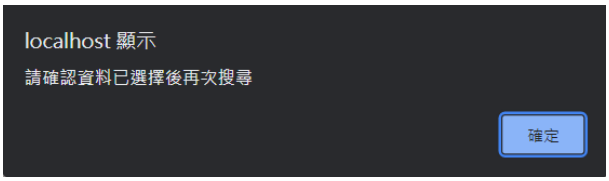
日期	進貨單編號	廠商名稱	製單人員		
2021-11-03	1000	惠亞瓷器股份有限公司	陳勳森	刪除單據	查詢明細
2021-10-12	1001	惠亞瓷器股份有限公司	謝裕民	刪除單據	查詢明細
2020-12-31	1002	惠亞瓷器股份有限公司	翁崇銘	刪除單據	查詢明細
2021-11-23	1003	現代瓷器股份有限公司	郭淑玲	刪除單據	查詢明細
2021-09-15	1004	現代瓷器股份有限公司	廖述宏	刪除單據	查詢明細

1 2 | 下一頁

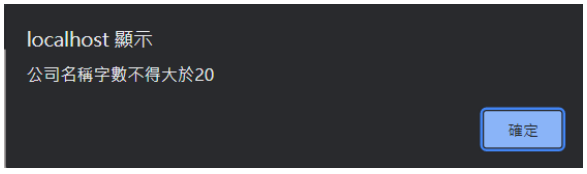
5.4 防呆機制

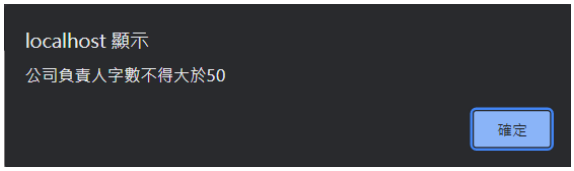
(i) 填寫資料不可為空

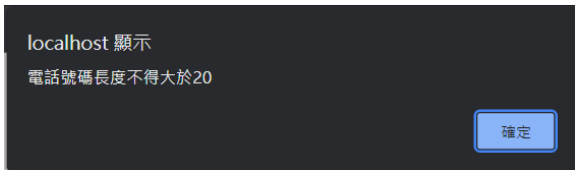




(ii) 資料大小超過限制





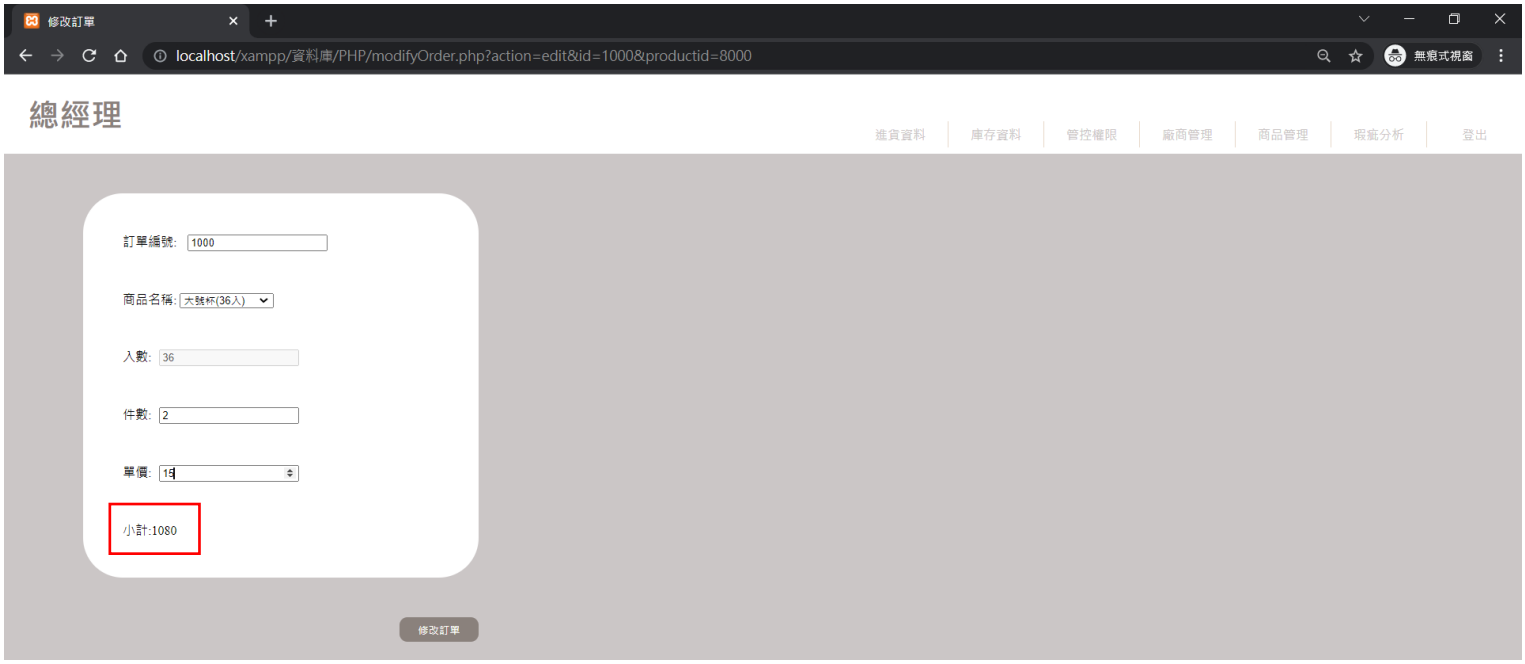


5.5 即時顯示

影片：

https://drive.google.com/file/d/1GvUvvxCTeh4a-KTyWyZIG0J8XWtQNUFD/view?usp=sharing

可以在畫面上即時計算出對應的金額:



5.6 訂單報銷單一次新增多筆

5.7 廠商綁定可選擇的商品

5.8 美編

本系統採用 css 來製作美編，主要的色彩為褐色及白色搭配